

Validity Is Not Difficulty: Hazard-Preserving Physical 3D World Compilation for Autonomous Driving

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Abstract

Driving-world reconstruction systems often conflate two fundamentally different causes of difficulty: a scene may be hard because it contains a legitimate hazardous actor, or invalid because its reconstructed geometry contradicts sensor evidence. Treating both cases as low confidence can obtain superficially safer worlds by deleting precisely the actors that matter. We introduce HARP-3D, a hazard-preserving physical 3D world compiler that separates appearance from collision geometry and artifact validity from behavioral hazard. The compiler aligns multi-frame Actor evidence in canonical coordinates and exposes four explicit surface actions—KEEP, PROJECT, COMPLETE, and UNKNOWN—without changing Actor identity, trajectory, or extent. A paired factorization objective holds artifact predictions invariant to legal hazard interventions and hazard predictions invariant to geometry artifacts. Finally, learned Actor residual distributions are combined with analytic trajectory-boundary geometry to produce a continuous, multi-horizon task-cost density and a runtime reliability surface that is monotone in both cost budget and horizon. On a frozen zero-shot Argoverse 2 cohort, ray-certified compilation eliminates paired free-space violations, reduces target LiDAR depth error from 0.207 to 0.154 m and symmetric Chamfer from 0.254 to 0.175 m, and retains all Actor and hazard state. A nearest-surface alternative attains 0.168 m Chamfer but leaves 84.9% matched-ray violations, exposing an explicit geometry–safety boundary. Remaining factorization and reliability values stay frozen until their milestones complete.

1. Introduction

Reconstructed driving worlds are increasingly used for perception analysis, counterfactual replay, and planning evaluation. Their geometry, however, is rarely a literal collision model. A rendered Gaussian may explain image appearance while occupying free space, duplicating an Actor shell, or

flickering across time. Conversely, a perfectly valid Actor may be difficult because it cuts in, brakes hard, or crosses the Ego path. A useful world compiler must distinguish these cases: *world validity is not world difficulty*.

This distinction matters because a generic confidence filter has an easy but scientifically undesirable failure mode. It can improve aggregate collision proxies by suppressing uncertain or nearby Actors. Such a world looks safer because legitimate hazards disappeared, not because its physical state became more faithful. We instead ask whether evidence-inconsistent geometry can be repaired while identity, lifecycle, trajectory, size, time-to-collision, and hazard events remain fixed.

We propose HARP-3D, a physical compilation layer between a reconstructed scene and downstream task queries. It represents a world as

$$\mathcal{W}_t = (\mathcal{G}_t^{\text{app}}, \mathcal{S}_t^{\text{phys}}, \mathcal{A}_t, \mathcal{P}_t, \mathbf{q}_t^{\text{val}}), \quad (1)$$

where appearance Gaussians and physical collision surfaces have separate semantics. The compiler uses multi-frame Actor alignment, ray free/occupied evidence, temporal support, and source provenance to select one of four local actions. Unsupported regions become UNKNOWN, never free by default, while Actors remain in the scene.

Our second component factorizes validity from hazard with paired counterfactuals. Changing a clean Actor from ordinary driving to a legal near-miss should not change its artifact score; injecting a duplicate shell or temporal pop should not change its hazard score. Our third component links physical state quality to downstream reliability. An Actor residual distribution is projected onto a candidate trajectory boundary, yielding a continuous cost whose conditional density supports arbitrary budgets and multiple horizons.

Our contributions are:

- a hazard-preserving physical compiler with explicit KEEP/PROJECT/COMPLETE/UNKNOWN semantics and separate appearance/collision state;
- a paired validity–hazard factorization protocol that measures cross-task leakage rather than relying on global

- 075 decorrelation;
 076 • a continuous task-cost density and monotone multi-
 077 horizon reliability interface, evaluated under a frozen
 078 nuScenes-to-ArgoVerse 2 zero-shot protocol.

079 2. Related Work

080 **Driving-world reconstruction.** Multi-sensor driving
 081 datasets such as nuScenes [1] and ArgoVerse 2 [7] provide
 082 calibrated imagery, LiDAR, ego pose, maps, and tracked
 083 Actors. Neural scene graphs model dynamic objects
 084 explicitly [4], while 3D Gaussian Splatting provides a
 085 high-quality appearance representation [2]. Our work does
 086 not reinterpret Gaussian opacity as collision occupancy.
 087 It adds a separate, queryable physical surface with sensor
 088 provenance.

089 **Uncertainty and reliability.** Deep ensembles expose use-
 090 ful empirical predictive disagreement [3], and conformal-
 091 ized quantile methods can turn residuals into calibrated
 092 marginal intervals under appropriate assumptions [5]. Driv-
 093 ing logs violate many simple exchangeability assumptions
 094 through scene correlation and domain shift. We there-
 095 fore report empirical calibration explicitly, preserve claim
 096 boundaries, and use monotone structure only for runtime
 097 consistency—not as a formal road-safety guarantee.

098 **Our distinction.** Prior confidence, occupancy, and
 099 reconstruction-quality signals often mix physical invalid-
 100 ity, visibility, and behavioral difficulty. HARP-3D instead
 101 defines separate evidence inputs, paired interventions, and
 102 failure denominators so that an improvement cannot be ob-
 103 tained by deleting hazardous Actors or relabeling unknown
 104 space as free.

105 3. Problem Formulation

106 Let Actor i have an immutable semantic state

$$107 A_i = (ID_i, X_{i,0:H}, D_i), \quad (2)$$

108 containing identity, trajectory, and dimensions. Appearance
 109 primitives may be associated with A_i , but collision queries
 110 consume only an Actor-local physical surface S_i^{phys} and its
 111 known/unknown mask.

112 We separate two random variables. Artifactness a_i de-
 113 pends on sensor and geometric evidence E_i , physical sur-
 114 face state, and provenance:

$$115 a_i = P(\text{artifact} \mid E_i, S_i^{\text{phys}}, P_i). \quad (3)$$

116 Hazardness h_i depends on Actor dynamics, map context,
 117 and a candidate Ego trajectory τ :

$$118 h_i = P(\text{hazard} \mid X_{i,0:H}, \tau, \mathcal{M}). \quad (4)$$

The goal is paired conditional invariance, not unconditional
 statistical independence. For the same clean Actor, a le-
 gal safe-to-hazard intervention should preserve a_i . For the
 same behavior, a clean-to-artifact intervention should pre-
 serve h_i .

For task reliability, let $n_{\tau,t}$ be the local boundary nor-
 mal, $d_{i,t}(\tau)$ the signed Actor clearance, and $\epsilon_{i,t}$ the Actor
 residual. We define

$$C(\tau, H) = \max_{i,t \leq H} \frac{|n_{\tau,t}^\top \epsilon_{i,t}|}{\max(|d_{i,t}(\tau)|, \epsilon)}. \quad (5)$$

The runtime interface returns $R(\tau, H, b) = P(C \leq b)$ and
 enforces $\partial R / \partial b \geq 0$ and $\partial R / \partial H \leq 0$.

4. Method

4.1. Actor-Preserving Physical Compilation

For each Actor, we transform LiDAR endpoints, rays, and
 associated primitives into a canonical Actor frame,

$$x_{i,t}^{\text{actor}} = T_{\text{actor},t}^{-1} T_{\text{ego},t} T_{\text{sensor}} x_t^{\text{sensor}}. \quad (6)$$

The fused representation stores surface support, observed-
 free intervals, temporal counts, local normals, viewing-
 direction diversity, and provenance. Compilation is deter-
 ministic and local:

- KEEP: stable sensor-supported surface;
- PROJECT: a near-surface primitive contradicts observed
free space and is projected to supported geometry;
- COMPLETE: a local hole has sufficient multi-frame and
multi-view support;
- UNKNOWN: evidence is missing, contradictory, or insuf-
ficient.

Unknown is a first-class state and never silently becomes
 free. All four actions preserve $(ID_i, X_{i,0:H}, D_i)$. Appear-
 ance primitives may attach to the repaired surface using cen-
 ter distance, covariance-normal alignment, depth, and visi-
 bility, but physical queries do not read Gaussian opacity.

4.2. Validity–Hazard Factorization

The validity encoder receives ray contradiction, surface
 residual, temporal support, provenance, visibility, and local
 geometry. The hazard encoder receives relative pose and
 velocity, TTC, route relation, map context, and interaction
 state. Two supervised heads are augmented with paired con-
 sistency losses:

$$\mathcal{L} = \mathcal{L}_{\text{artifact}} + \mathcal{L}_{\text{hazard}} + \lambda_{\text{pair}} \mathcal{L}_{\text{pair}} + \lambda_{\text{leak}} \mathcal{L}_{\text{leak}}. \quad (7)$$

Pairs span a four-quadrant atlas: valid-safe, valid-hazard,
 artifact-safe, and artifact-hazard. We measure hazard-from-
 validity and artifact-from-hazard probes in addition to task
 AUROC/AUPRC.

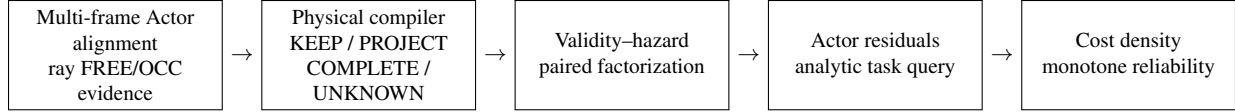


Figure 1. HARP-3D compiles sensor evidence into an Actor-preserving physical world, separates validity from hazard, and exposes task-conditioned reliability. Appearance Gaussians never serve directly as collision occupancy.

4.3. Actor Residuals and Boundary-Cost Density

For Actor i and horizon t , a low-capacity head predicts a 2D residual distribution $(\mu_{i,t}, \Sigma_{i,t})$. Analytic projection onto $n_{\tau,t}$ gives mean $n^\top \mu$ and variance $n^\top \Sigma n$. Equation 5 converts these residuals and deterministic clearance into a continuous trajectory cost.

We model $y = \log(1+C)$ with a small conditional Gaussian mixture,

$$p_\theta(y | z^{\text{val}}, z^{\text{haz}}, \tau, H) = \sum_k \pi_k \mathcal{N}(y; \mu_k, \sigma_k^2), \quad (8)$$

which exposes a closed-form marginal CDF for any budget. A low-rank, input-conditioned dependence model combines the frozen marginals across horizons. This factorization separates learned Actor uncertainty from analytic task geometry.

4.4. Monotone Runtime Surface and SceneIR

The final runtime object is a frozen grid over horizon and cost budget with monotone interpolation. Budget CDF values are non-decreasing; prefix-horizon reliability is non-increasing. Groupwise isotonic maps may calibrate predefined conditions on a disjoint source fold, but do not change these structural constraints.

SceneIR stores Actor identities and transforms, physical surfaces, known/unknown masks, provenance, and validity fields. The AV2 adapter performs only coordinate, unit, timestamp, category, and sensor-opportunity normalization. It does not fine-tune, recalibrate, or change thresholds using AV2 quality.

5. Experiments

Data and protocol. We train and select models only on nuScenes. Argoverse 2 Sensor validation logs form a zero-shot external test. Before reading any method output, we sort the 150 validation UUIDs and select every fifth log, yielding 20 quantitative and 10 qualitative logs. No AV2 fine-tuning, post-hoc calibration, threshold selection, or failed-scene deletion is allowed. Final nuScenes and AV2 cohorts are evaluated exactly once after the method freezes.

Baselines. For physical compilation we compare naive reconstruction, a legacy validity filter, V6.7 inward-ray repair,

and HARP-3D. For factorization we compare a shared encoder, a two-head shared trunk, independent encoders, and paired factorization. Reliability baselines include a scalar confidence, independent horizon classifiers, a continuous marginal density, and the full joint multi-horizon surface.

Metrics. Physical metrics include free-space violation, ray termination, point-to-surface error, LiDAR depth consistency, temporal jitter, ghost components, and surface completeness. Hazard preservation reports Actor/ID/lifecycle retention, trajectory and TTC shift, and event-count change. Factorization reports artifact/hazard AUROC and AUPRC, clean-hazard false-artifact rate, paired prediction shift, and cross-probe leakage. Reliability reports log likelihood, integrated Brier score, calibration error, Spearman correlation, monotonic violations, and runtime.

Zero-shot Actor-local surface evidence. Following the dynamic-Actor canonicalization boundary used by NeuRAD [6], we transform ego-frame AV2 LiDAR points inside each annotated cuboid into its Actor frame. Unlike appearance-oriented seed initialization, we never mirror points: every collision surfel must retain sensor and temporal support. We fuse two of every three frames at 0.12 m and reserve the remaining frames as unseen target geometry. Across 1,046 Actors and 1.075M stable surfels, fusion reduces the mean target-to-surface distance from 0.864 m to 0.131 m and raises recall at 0.20 m from 26.16% to 85.72% (Table 1).

The paired corruption atlas contains clean, observed-free ghost, duplicate-shell, temporal-flicker, and supported-hole probes without changing Actor identity, trajectory, size, or hazard. It yields 100.0% artifact recall, 0.00% clean-hazard false-artifact rate, and 100.0% Actor/hazard retention. These perfect paired scores verify the deterministic compiler contract and input separation; they are not reported as natural-artifact generalization.

Four-action physical compilation. We next freeze one single-frame query per Actor and reserve all later held-out frames as target-only geometry. The compiler reads only the build surface and query evidence. A nearest-canonical projection initially reduces Chamfer to 0.168 m,

AV2 Actor surface	Dist. (m) ↓	Recall (%) ↑
Single observed frame	0.864	26.16
Multi-frame canonical surfels	0.131	85.72

Table 1. Zero-shot held-out LiDAR support on the frozen 20-log AV2 quantitative cohort (1,046 Actors, including 241 hazardous Actors). The comparison uses real sensor points; paired synthetic corruptions are evaluated separately as a compiler contract.

Surface	Free (%) ↓	Recall (%) ↑	CD (m) ↓
Single-frame input	100.0	53.36	0.254
Nearest canonical	84.91	62.60	0.168
Ray-certified	0.00	59.56	0.175

Table 2. Frozen zero-shot AV2 quantitative result. Nearest-surface projection fits geometry best but violates matched-ray free space; ray-certified projection accepts a small Chamfer cost. Later held-out frames are target-only.

but ray-specific evaluation reveals 84.91% early terminations in the paired observed-free interval. Our final operator is ray-certified: a primitive with direct LiDAR provenance projects to its matched observed termination; without one it becomes UNKNOWN. Across 433 quantitative Actors (147 hazardous), this reduces paired free-space violations from 100.0% to 0.00%, target depth error from 0.207 m to 0.154 m, and Chamfer from 0.254 m to 0.175 m, while raising ray-termination consistency from 56.94% to 64.38% (Table 2). Actor identity, trajectory, extent, and hazard labels are retained for 100.0% of Actors. The zero free-space rate is a paired contract result; depth, ray termination, and Chamfer use independent target-only LiDAR.

Qualitative protocol. Main-paper and supplement scenes are frozen from metadata and registered failure strata before rendering method outputs. Panels show RGB, LiDAR, physical surface, ray evidence, compiler action, and the repaired world. We report all frozen cases, including failures, rather than selecting only visually favorable examples.

6. Limitations

HARP-3D is not a formal guarantee of real-road safety. Sparse or one-sided sensing can leave large regions unknown, and deterministic completion is intentionally conservative. Actor cuboids and tracked identities may themselves be wrong. Empirical isotonic calibration under scene correlation does not provide distribution-free conditional coverage, and nuScenes-to-AV2 transfer may fail through sensor opportunity, interface, representation, dynamics, or calibration shift. The current physical compiler targets Actor-local surfaces; static-scene topology and deformable Actors remain incomplete. Finally, preservation of iden-

tity, trajectory, and size guarantees only hazard functions that depend on those variables; appearance-dependent human judgments may still change.

7. Conclusion

We formulate driving-world compilation around a simple boundary: validity is not difficulty. HARP-3D repairs evidence-inconsistent physical geometry while preserving legitimate hazardous Actors, factorizes artifactness from hazardness with paired interventions, and exposes downstream reliability through a continuous multi-horizon cost distribution. The resulting SceneIR keeps appearance, collision state, unknown space, provenance, and task reliability explicit. The remaining work is empirical: completing the canonical-surface milestones, executing the frozen zero-shot protocol, and replacing all draft placeholders without expanding the claim boundary.

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Method	Free-space ↓	Surface error ↓	Actor retention ↑	Clean-hazard FA ↓	Brier ↓	AV2 zero-shot ↑
Naive reconstruction	TBD	TBD	TBD	TBD	TBD	TBD
V6.7 inward-ray	TBD	TBD	TBD	TBD	TBD	TBD
HARP-3D	TBD	TBD	TBD	TBD	TBD	TBD

Table 3. Paper-facing result contract. Values remain TBD until the corresponding frozen milestone completes. Metrics will be split into full physical, factorization, and reliability tables in the results draft.